ATE–01 / MCU–01
ESP32–S3
Virtual ATE Simulator

– Test Result Simulation
– ATE_READY
– TEST_BUSY / TEST_DONE
– RESULT_BIN
– ATE_FAULT
– HEARTBEAT

Modbus RTU Master

RS–485 MASTER

Virtual ATE Status / Result Data to PLC
– ATE_READY
– TEST_BUSY
– TEST_DONE
– RESULT_BIN
– ATE_FAULT
– HEARTBEAT

Written by ESP32 Master
to PLC registers/coils

RS–485 (2–Wire Half–Duplex)
Modbus RTU
9600 bps, 8N1

ESP32–S3 : Modbus RTU Master
PLC–01 : Modbus RTU Slave, ID = 1

PLC Command / Request Data to Virtual ATE
– TEST_REQ
– TEST_ACK
– RESET_REQ

Read by ESP32 Master from PLC registers/coils

SCADA–01
LabVIEW SCADA / HMI

– Auto / Manual Status
– Station Step Monitoring
– Alarm History
– Production / Bin Count Log
– Cycle Time Trend
– Start / Stop / Reset Request

COMMAND REQUEST ONLY
PLC INTERLOCK APPROVAL REQUIRED

RS–232C
XGT Protocol
9600 bps, 8N1,
No Flow Control

PLC Status Data
– Auto / Manual Mode
– Station Step
– Alarm Code / Alarm State
– ATE Communication Status
– PASS / FAIL / RETEST Counter
– Cycle Time
– Heartbeat / PLC Alive

HMI Command Request
– Start Request
– Stop Request
– Reset Request
– Auto / Manual Mode Request
– Recipe / Parameter Request

PLC validates all command requests
before any physical output is energized.

[CONTROL AUTHORITY]

1. PLC–01 is the final authority for:
   – Sequence execution
   – Safety interlock
   – Alarm handling
   – Conveyor / Clamp / Gate / Lamp / Buzzer outputs
2. ESP32 Virtual ATE provides only:
   – Test status
   – Test completion
   – Result Bin
   – Heartbeat / Fault status
3. LabVIEW SCADA provides:
   – Monitoring
   – Logging
   – Operator command request
4. Neither ESP32 nor LabVIEW may directly energize field actuators.

[DESIGN NOTES]
[1] PLC–01 is the only controller with final authority over field actuator outputs.
[2] ESP32–S3 emulates a Virtual ATE and acts as Modbus RTU Master on the RS–485 network.
[3] Logical signal direction is independent of physical Modbus Master/Slave transaction direction.
[4] LabVIEW SCADA issues command requests only; PLC safety interlocks determine execution permission.

PLC–01
LS XBC–DR32H
Main Cell Controller

– Auto / Manual Sequence
– Safety Interlock & Alarm
– ATE Handshake Control
– Conveyor / Clamp / Gate Control
– Production Counter / Cycle Time

FINAL AUTHORITY FOR ACTUATOR OUTPUTS

PLC DIRECT I/O CONTROL
Safety / Interlock / Alarm Protected

COMMUNICATION NODE SETTINGS

Node	Tag	Interface	Protocol	Role
Virtual ATE	ATE–01 / MCU–01	Rs–485	Modbus RTU Master	Test result simulation
Main Controller	PLC–01	RS–485 / RS–232C	Modbus RTU Slave ID 1 / XGT	Sequence, safety, sorting
SCADA/HMI	SCADA–01	RS–232C	XGT Protocol	Monitoring, logging, request
Field Device	FD–01	Direct I/O	24VDC I/O	Sensor/actuator interface

RS–485: 9600 bps, 8N1, Half–Duplex
RS–232C: 9600 bps, 8N1, No Flow Control

[COMMUNICATION FAULT POLICY]

ATE Heartbeat Timeout
→ New test request inhibited
→ Active cycle held safely
→ Alarm: ALM_ATE_COMM_TIMEOUT

Invalid Result Bin
→ No automatic bin discharge
→ Hold product at safe position
→ Alarm: ALM_INVALID_BIN

LabVIEW Communication Loss
→ PLC continues safe autonomous control
→ PLC sets ALM_SCADA_COMM_LOSS
→ SCADA monitoring and logging unavailable
→ Field safety is not affected

LEGEND
Physical Communication Link
Logical Data Flow
Control Authority / Direct I/O Control

FIELD DEVICES
PACKAGE FINAL TEST & MULTI-BIN SORTING CELL

INPUT DEVICES
- ES–01 Emergency Stop (NC)
- PE–01 Infeed Presence Sensor
- PE–02 Test Position Sensor
- PE–03 Outfeed Confirm Sensor
- DR–01 Guard Door Interlock

OUTPUT DEVICES
- MTR–01–RUN Conveyor Motor Run
- MTR–01–DIR Conveyor Motor Direction
- SOL–01 Test Clamp
- SOL–02 PASS Gate
- SOL–03 FAIL Gate
- SOL–04 RETEST Gate
- LT–01–G Tower Lamp Green
- LT–01–Y Tower Lamp Yellow
- LT–01–R Tower Lamp Red
- BZ–01 Buzzer

VIRTUAL ATE HANDSHAKE SUMMARY

No.	Signal	Logical Direction	PLC Meaning
1	TEST_REQ	PLC → ATE	Package unit is positioned and ready for test. PLC requests test execution.
2	ATE_READY	ATE → PLC	Virtual ATE is available and ready to accept a test request.
3	TEST_BUSY	ATE → PLC	Test is in progress. PLC inhibits product movement and sorting operation.
4	TEST_DONE	ATE → PLC	Test execution is complete. PLC proceeds to result validation.
5	RESULT_BIN	ATE → PLC	Test result classification: PASS, FAIL, or RETEST.
6	TEST_ACK	PLC → ATE	PLC confirms successful receipt and validation of the test result.
7	ATE_FAULT	ATE → PLC	Virtual ATE reports a test–system fault or simulated equipment fault.
8	HEARTBEAT	ATE → PLC	Virtual ATE communication–alive indication monitored by PLC.
9	RESET_REQ	PLC → ATE	PLC requests reset of the Virtual ATE fault state.

Logical direction represents data ownership.
Physical Modbus RTU transactions are initiated by ESP32 Master.

NO.	DATE	REVISION
A	2026–08–06	INITIAL RELEASE

DWG TITLE
EDS Wafer Map Linked Package
Final Test and Multi–Bin Sorting Cell

Control and Communication Architecture

DRAWN BY: J. H. KO
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